

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
O-LEVEL · MATHEMATICS

MATHEMATICS 4004/2

PAPER 2 November 2020

2 hours 30 minutes

PAPER 2 — STRUCTURED QUESTIONS — CALCULATOR ALLOWED

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers. This script is unique to November 2020 4004/2.

Additional materials: Mathematical tables or a non-programmable calculator may be used.

INSTRUCTIONS: Section A: answer all questions (52 marks). Section B: answer any four of the seven questions (48 marks).

SECTION A (52 marks) — Answer ALL questions

1. (a) Make t the subject of $v = u + at$. [3] <<
>> (b) Simplify $(3x/4) + (x/6)$. [3] <<
>> (c) Solve $2/x = 5/(12)$. [4] **[10]**

2. $AB \parallel CD$ with transversal EF in a street plan of Gweru. <<
>> (a) Angle $APQ = 61^\circ$. State the corresponding angle at Q , with a reason. [2] <<
>> (b) Find the co-interior angle to 61° . [2] <<
>> (c) A triangle has angles 61° , 42° and x° . Find x . [2] <<
>> (d) Classify the triangle (acute / right / obtuse). [2] **[8]**

3. A rectangular garden in Mkoba measures 12 m by 11 m. A path of width 1 m runs all around the outside. <<
>> (a) Area of the garden. [2] <<
>> (b) Area of the path. [3] <<
>> (c) A circular bed of radius 7 m is later dug. Area of the bed, $\pi = 22/7$. [3] **[8]**

4. 30 Form 4 pupils at Moleli scored: <<
>> 4 (6 pupils), 3 (6 pupils), 2 (4 pupils), 3 (6 pupils), 6 (8 pupils). <<
>> (a) Modal mark. [1] <<
>> (b) Mean mark. [4] <<
>> (c) Range. [2] <<
>> (d) One extra pupil scores 6. Effect on the mean? [2] **[9]**

5. (a) Using ruler and compasses only, construct triangle ABC with $AB = 6$ cm, $BC = 5$ cm and $AC = 8$ cm. [4] <<
>> (b) Construct the perpendicular bisector of BC . [2] <<
>> (c) Mark the circumcentre and describe its property. [2] **[8]**

6. Line $L: y = 2x + -2$. <<
>> (a) Gradient and y -intercept. [2] <<
>> (b) Coordinates where L meets the x -axis. [2] <<
>> (c) Show $A(0, -2)$ lies on L . [2] <<
>> (d) Equation of the line parallel to L through $(0, 4)$. [2] **[8]**

SECTION B (48 marks) — Answer any FOUR of the seven questions

- 7.** $OA = a = (3 ; 3)$, $OB = b = (1 ; 6)$. M midpoint of AB. <<
>>(a) Vector AB. [3]<<
>>(b) $|AB|$. [3]<<
>>(c) Position vector OM. [3]<<
>>(d) Show $AM = MB$. [3] **[12]**
- 8.** A kombi leaves Gweru at 08:00 and travels 180 km at constant speed, arriving after 2 hours. It stops 30 minutes, then returns at 80 km/h. <<
>>(a) Outward speed. [3]<<
>>(b) Time for the return. [3]<<
>>(c) Sketch a distance–time graph, labelling axes. [4]<<
>>(d) Average speed for the whole trip including the stop. [2] **[12]**
- 9.** 60 pupils at Moleli sat a test (0–50). An ogive is drawn. <<
>>(a) How to read the median from the ogive. [3]<<
>>(b) Median estimated as 30. Meaning? [2]<<
>>(c) How to find the IQR from the graph. [4]<<
>>(d) Why IQR is better than the range here. [3] **[12]**
- 10.** (a) Solve $x^2 - 11x + 24 = 0$ by factorisation. [4]<<
>>(b) A rectangular plot in Gweru is 8 m longer than it is wide. If the area is 33 m^2 , form and solve an equation for the width w . [5]<<
>>(c) Sketch $y = (x - 3)(x - 8)$, showing intercepts. [3] **[12]**
- 11.** (a) y varies directly as x . When $x = 4$, $y = 36$. Find y when $x = 10$. [4]<<
>>(b) z varies inversely as x . When $x = 3$, $z = 8$. Find z when $x = 6$. [4]<<
>>(c) A farmer near Gweru has at most 30 hours. Maize takes 2 h/ha, vegetables 1 h/ha. Inequality for hectares m and v . [4] **[12]**
- 12.** A cellphone tower is observed from A on level ground in Gweru. Angle of elevation of the top is 30° . Distance A to the foot is 20 m. <<
>>(a) Height of the object. Use $\tan 30^\circ = 1/\sqrt{3}$. [4]<<
>>(b) From B, 20 m due East of A, the bearing of the foot is 300° . Sketch. [3]<<
>>(c) A bird sits halfway up. Angle of elevation from A. [5] **[12]**
- 13.** O is the centre. A, B, C on the circumference. AB is a diameter. Angle $BAC = 28^\circ$. <<
>>(a) Angle ACB. Reason. [3]<<
>>(b) Angle BOC (centre, same arc BC). [3]<<
>>(c) D on the remaining circumference. Angle BDC. Reason. [3]<<
>>(d) State the angle-in-a-semicircle theorem. [3] **[12]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).